NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state three hypotheses (H1 salience, H2 framing, H3 tokenizer cost) and report what was found: of ten pre-specified political concepts only parliament differs (uncorrected); the exploratory religious gap holds in readers' letters, in the same years and under the same editor (Jabavu), while the editor change within Isigidimi is not distinguishable from random words; political vocabulary followed the times; and the negative results (framing is weak and non-specific; the directional lean survives correction in three of five settings, but in at most one without land, and a neutral control leans further; the lean rests on umhlaba, which the two papers use in different senses). Claims about meaning are scoped as unreviewed by a first-language speaker.

   Guidelines:

   - The answer NA means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Discussion ends with limitations (two newspapers, one window, 1886 Isigidimi editions missing, an incomplete control-by-editor design, a heuristic letter split, a modern-news domain mismatch, unreviewed stems and glosses; the Results flag the one-model tokenizer hypothesis, and Table S3 the era-confounded held-out windows) and states the detection floor of the framing test.

   Guidelines:

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   - The authors are encouraged to create a separate "Limitations" section in their paper.
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   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
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3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [NA]

   Justification: The paper has no theoretical results.

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   - The answer NA means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
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   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
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4. **Experimental Result Reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The Methods give the window, the date-matched subsampling, concept stems, model and hyper-parameter choices, the permutation null, the matched-set tests and the tokenizer conditions. Anchor lists are summarised in Table S4; word-joining and tokenizer rewrite rules are in the submitted code and README.txt, which gives the run order; seeds are fixed.

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- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
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5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: The code is submitted with requirements.txt and README.txt (exact run order). The newspaper texts are the public IsiXhosa Intellectual Traditions archive: the course-provided zip is read by src/01_prepare_data.py and the same texts can be downloaded from the archive DOIs listed in README.txt (Schoots et al., 2023a); the modern text is public (MasakhaNEWS) and README.txt gives the download command.

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- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental Setting/Details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Hyper-parameters were chosen on text outside the test window (Isigidimi 1876-83, Imvo 1889-94) by seed stability over four grid rounds (34 settings; Table S8 and outputs/tuning_grid.csv), the classifier uses issue-grouped nested cross-validation, and the subsampling and permutation procedures are described.

Guidelines:

- The answer NA means that the paper does not include experiments.

- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment Statistical Significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Salience has issue-level bootstrap CIs; framing uses a permutation null (50 permutations) and 2000 frequency-matched word sets; tokenizer changes have bootstrap CIs (within about 0.5%, stated in the Results). The 15 salience tests are uncorrected and none survives Bonferroni ($p < 0.003$); the framing tests are reported uncorrected, except the directional test, which survives Bonferroni over five settings ($p < 0.01$) in three, but in at most one without land, and a neutral control leans further; the paper rests its argument on robustness and corroboration. The held-out comparison is era-confounded and is presented as an era-sensitivity check, with within-paper drift (including vote's large drift) as calibration; the classifier was trained on the test window, so religion is exploratory. The same-editor contrasts are tested against matched sets as well as bootstrap CIs; the letters/rest comparison and the senses of umhlaba use issue-bootstrap CIs only; the letter detector was checked by hand on 30 random detections (28 genuine), and its recall was not measured.

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   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
   - If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments Compute Resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: All experiments ran on CPU only (no GPU) in roughly three hours in total, including tuning rounds and the earlier framing design that was replaced; README.txt gives the four-core setup.

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   - The answer NA means that the paper does not include experiments.
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   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work uses public-domain historical newspapers and public modern news text, collects no data from people, and flags that interpretation needs first-language-speaker review. To our knowledge it conforms to the Code of Ethics.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Discussion covers fairness implications (higher tokenizer cost for historical isiXhosa, worst for the African-language model; the case against replacing the source with a normalised copy) and the risk of over-reading statistical shifts as cultural meaning.

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Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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Justification: New assets are the analysis code and derived word lists (anchors, word-joining rules, the speaker-check sheet), documented in README.txt; no new text corpus is released.

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